Caffeine Consumption II: Lab Session

Lab sessions are boring. You'd much rather be outside drinking some Mother™. You just can't get enough of that invigorating buzz and refreshing taste. Why are you even still here? They're not even getting you to do useful problems; nobody in the history of ever has actually needed to code a max flow. Screw this. You're going to go back to your room and get some sweet Mother™.

Bernard is off somewhere else preparing the selection exam, so you can slip out of the lab unnoticed. Unfortunately, the crafty devil has sent all the tutors to stand guard outside. Evading them will be a true test of your ninja skills and your devotion to Mother™.

The path back to the college is rectangular and bordered by buildings on the north and south. The tutors stand at specific locations on the path and do not move. You must find a path between buildings which stays *strictly more* than 100 metres away from a tutor (this is the maximum distance at which they can spot you, as you learned at dinner last night).

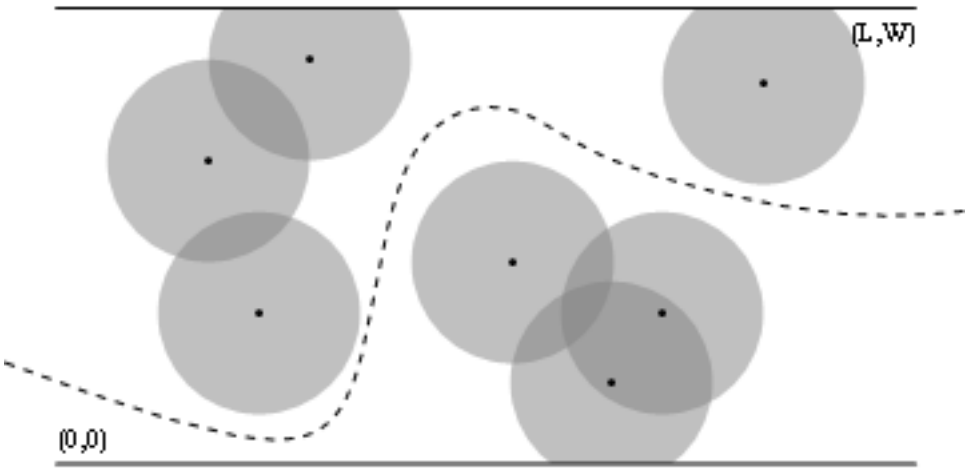


Figure 1:

The above diagram shows an example of this. Representing the path as an L by W co-ordinate system, you start at x=0 and must make your way to x=W. The dotted line represents one possible path from left to right.

Of course, Bernard may have arranged the tutors so that they are impossible to evade. In this case, you can bribe individual tutors with cans of Mother™. (They will be so enamored by the prospect of sculling down a delicious can of Mother™ that they will not report back to Bernard.) However, sharing your personal supply of Mother™ is a last resort, and should be done as little as possible.

Actually, how many tutors *will* you have to bribe?

Input

The first line of input will contain the integers L, W and N, separated by spaces. These respectively denote the length and width of the path to the college, and the number of tutors. You are guaranteed that 1 <= L, W <= 50,000, and 1 <= N <= 250.

Each of the following lines contains a pair of integers x and y (0 <= x <= L, 0 <= y <= W) - the location of one tutor.

Output

Your output file should consist of a single integer: the minimum number of tutors who must be bribed (with Mother™) on your way back to the college.

Sample Input

```
130 340 5
10 50
130 130
70 170
0 180
60 260
```

Sample Output

```
1
```

Scoring

The score for each input file will be 100% if the correct answer is written to the output file and 0% otherwise. Used Mother™ cans may be exchanged for extra points.